

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



**ST. ANDREW'S JUNIOR COLLEGE
2014 JC2 Preliminary Examinations**

H2 BIOLOGY

9648/2

Paper 2: Core

Tuesday

2nd September 2014

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/12
3	/10
4	/12
5	/13
6	/11
7	/12
Total	/80

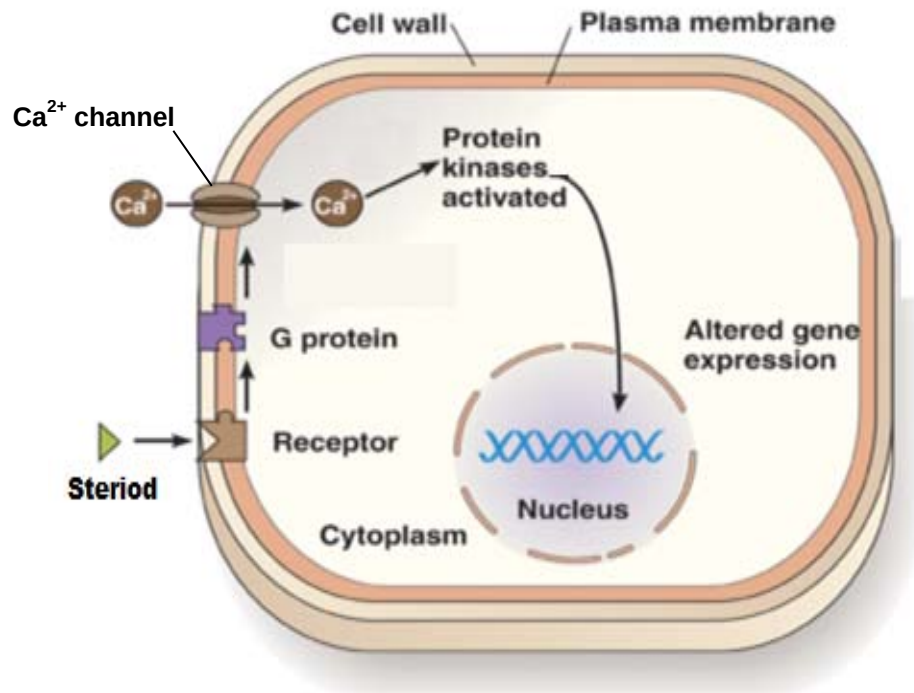
This document consists of **20** printed pages.

[Turn over

Section A

Answer **all** questions.

1(a) Fig. 1.1 shows the plant cell transmembrane cell signalling mechanism which is comparatively less well-studied than that of the animal cell.



Source: Adapted from <https://plantsrock.wikispaces.com/file/view/Picture1.png/30513514/409x329/Picture1.png>

Fig 1.1

(i) State the main stages of *cell signalling*.

.....
[1]

(ii) With reference to **Fig. 1.1**, describe the signal transduction within the plant cell upon steroid binding.

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.....[4]

According to Michael Hothorn of the Salk Institute, the transmembrane signalling mechanism used by plants differ from those used by animals. A small steroid molecule on the outside of the plant cell assists in the transmembrane signalling process. By contrast, this sort of molecule and its receptor is generally located intracellularly in animal cells.

(iii) Explain how such signal molecules enter the cytoplasm of the animal cell.

.....

.....[1]

Beta cells in the Islets of Langerhans in the pancreas secrete peptide hormone, insulin into the blood stream. The production of functional insulin and its secretion is done with the help of vesicles.

Proteins embedded in the membranes of vesicles have different functions.

- COPII and v-SNARE proteins are known as 'address proteins'.
- Vesicles that transport insulin from the rough endoplasmic reticulum to the Golgi body are coated with COPII proteins.
- Vesicles that transport insulin from the Golgi body to the cell surface membrane are coated with v-SNARE proteins.

(b) Outline how the vesicles are moved from one cellular location to another and suggest how these 'address proteins' ensure that vesicles are transported to the correct locations within the cell.

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.....[2]

Insulin travels via the bloodstream to bind to the insulin receptors embedded in the plasma membrane on target cells. The plasma membrane can be described by the fluid mosaic model.

(c) Describe what is meant by the fluid mosaic model.

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.....[2]

[Q1 Total: 10]

2(a). A strain of the bacterium, *Streptococcus pneumonia*, has been found which is resistant to the antibiotic vancomycin. This vancomycin resistance is caused by a mutation of a gene coding for an enzyme, S.

(i) Suggest and briefly explain 2 different ways in which a mutation of the gene may still result in the production of a functional enzyme.

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.....
.....[2]

(ii) Explain briefly how a mutation of the gene may result in an enzyme with reduced activity.

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.....[2½]

(iii) A random mutation resulting in resistance to vancomycin was found to occur on a plasmid. Explain two ways how the resistance to vancomycin may spread naturally amongst bacteria.

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.....[3]

The control of the expression of the lac operon genes, which allow uptake and digestion of lactose in the bacterium *Escherichia coli*, is well known.

Fig.2.1 shows the arrangement of the elements of the *lac* operon.

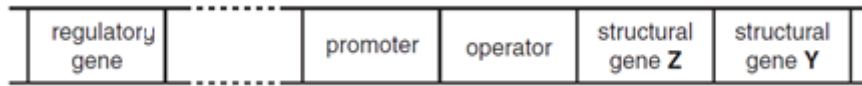


Fig.2.1

(b)(i) Distinguish between structural and regulatory gene.

.....
[1]

(ii) Describe how genes Z and Y are switched on in bacteria that are moved to a nutrient medium that contains lactose.

.....

[3½]

[Q2 Total: 12]

3. The following is an excerpt taken from the Health Promotion Board's website .

"Influenza is a respiratory illness which is highly contagious. It is a serious condition because the infection, at times, can lead to complications and even death. The best way to prevent the flu is by getting a flu vaccination each year."

(a) Describe the structural features of the influenza genome.

.....
.....
.....[1½]

(b) Suggest why the influenza vaccine has to be changed each year.

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.....[3½]

Tamiflu® is an antiviral drug that can be used to treat influenza patients.

(c) State why a doctor would not prescribe antibiotics to treat influenza.

.....
.....[½]

Tamiflu[®] is a neuraminidase inhibitor. **Fig.3.1** shows the structures of Tamiflu[®] and sialic acid.

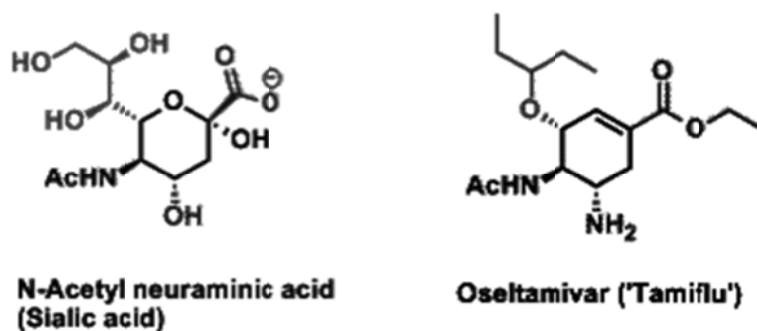


Fig.3.1

(d) With reference to **Fig.3.1**, explain how Tamiflu[®] could be administered to inhibit neuraminidase.

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[2½]

(e) Suggest how Tamiflu[®] could help to reduce the spread of influenza.

.....

[2]

[Q3 Total: 10]

4. Neuregulin 1 (NRG1) gene is a potential gene which contributes to the development of breast cancers. Results of a study (Chua *et al.*, 2009) on DNA methylation and gene expression of the NRG1 gene in different breast cell lines were shown below in **Fig. 4.1** and **4.2**.

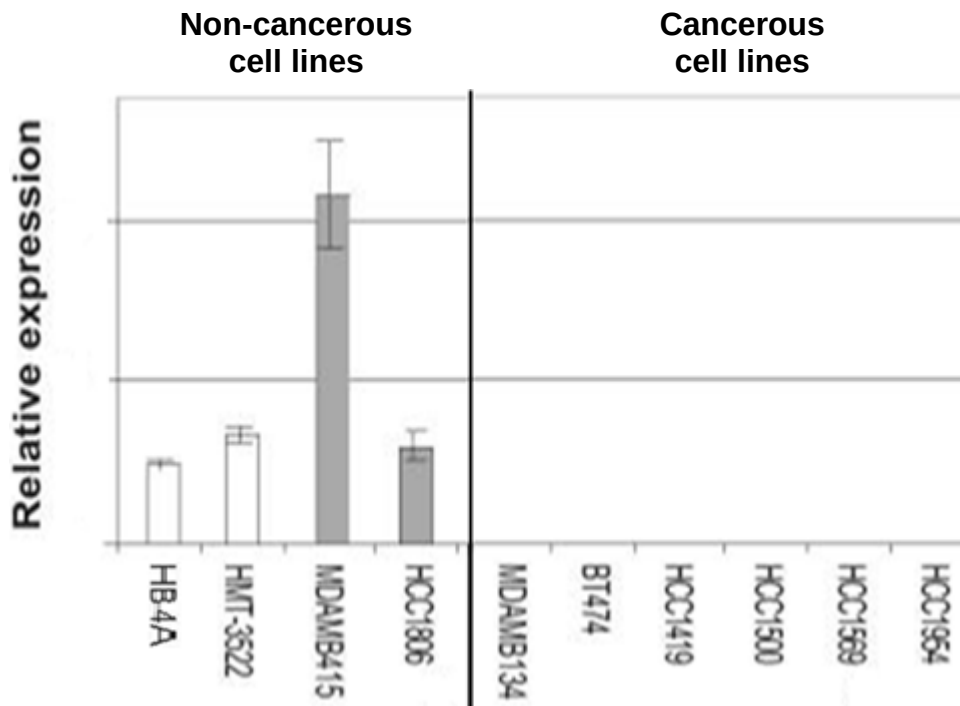


Fig. 4.1. Relative expression of NRG1 gene in various cell lines. Names of cell lines are reflected on the x-axis.

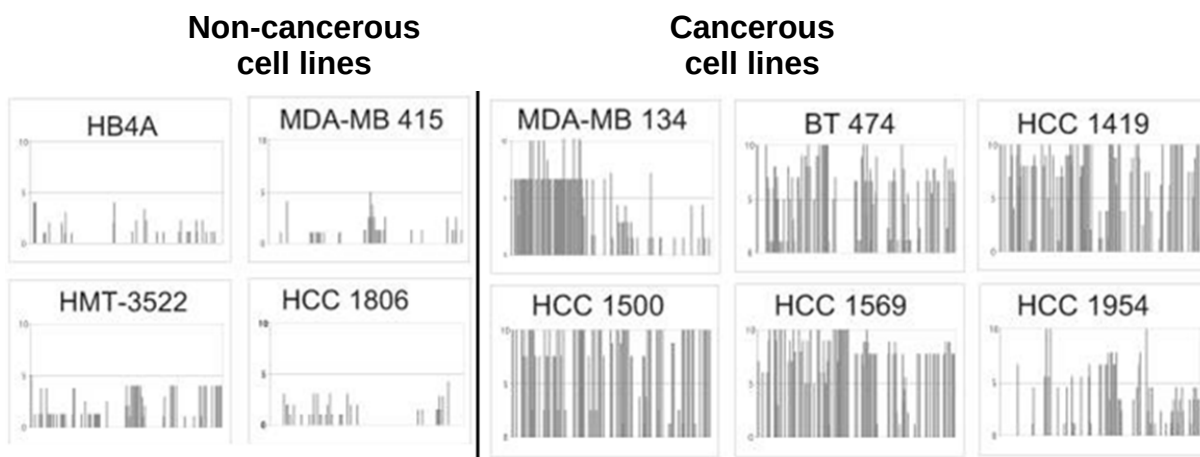


Fig. 4.2. Extent of DNA methylation of NRG1 gene, as represented by the quantity of vertical lines present, in the various cell lines.

(a) With reference to **Fig. 4.1** and **4.2**, explain the relationship between the extent of DNA methylation and NRG1 gene expression.

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[2½]

(b) The expression of some other genes may be regulated negatively at the post-translational level. List two examples of such regulation.

.....
[1]

(c) Cancer is a multi-step process involving the mutations of both proto-oncogenes and tumour suppressor genes. With reference to gene amplification, explain how proto-oncogenes can be converted to oncogenes.

.....

[1½]

(d) Cytoplasmic mRNA of all expressed genes was isolated from the BT474 cell line. This RNA was mixed with a purified, denatured DNA fragment that carried a large intron of a housekeeping gene (a gene expressed in essentially all cells), and the mixture was incubated for 12 hours under re-naturation conditions.

Explain whether RNA-DNA duplexes would be formed in this experiment.

.....
[1]

(e) The end-replication problem explains why normal somatic cells are limited in the number of mitotic divisions before they die out.

(i) Outline the end-replication problem in eukaryotic chromosomes.

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[3]

(ii) Explain how cells of the BT474 cell line overcome the end-replication problem.

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.....[3]

[Q4 Total: 12]

5. In a particular species of corn harvested from a vegetable farm, the allele *A* allows the deposition of anthocyanin (blue) pigment in the kernels (seeds), while *aa* plants have yellow kernels. At a second gene locus, *B*– produces smooth kernels, while *bb* kernels are wrinkled.

A plant, heterozygous for blue smooth kernels was crossed to a plant with yellow wrinkled kernels. The progeny consisted of 1447 blue smooth, 169 blue wrinkled, 186 yellow smooth, and 1510 yellow wrinkled.

(a)(i) Define the term gene locus.

.....
[1]

(ii) Define the term heterozygous.

.....
[1]

A chi-squared test was carried out for this cross.

<i>Phenotypes</i>	<i>Observed number (O)</i>	<i>Expected number (E)</i>	$\frac{(O - E)^2}{E}$
Blue smooth	1447	828	462.75
Blue wrinkled	169	828	524.49
Yellow smooth	186	828	497.78
yellow wrinkled	1510	828	561.74
$\chi^2 = 2046.78$			

Table 1: A chi-squared table.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(b)(i) Use the calculated value of χ^2 of 2046.78, and the table of probabilities provided in **Table 1** to find the probability of the observed results departing significantly by chance from the expected ratio.

.....[1/2]

(ii) State what conclusions may be drawn from the probability found in (b)(i).

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.....
.....[1 1/2]

(c) Using the symbols provided, draw a genetic diagram for the above cross.

.....[4]

(d) Comment on the locations of these two genes' loci. Show your calculations clearly.

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.....[2]

(e) Honeybees were also observed to visit the corn plants for pollen. With reference to honeybees, explain how the environment may affect the phenotype.

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.....[3]

[Q5 Total: 13]

6. Palisade cells are plant cells found within the mesophyll in leaves of many plants, right below the upper epidermis and cuticle. They have both chloroplasts and mitochondria. Exchanges between a mitochondrion, a chloroplast and the cytoplasm surrounding them are shown in the **Fig. 6.1**

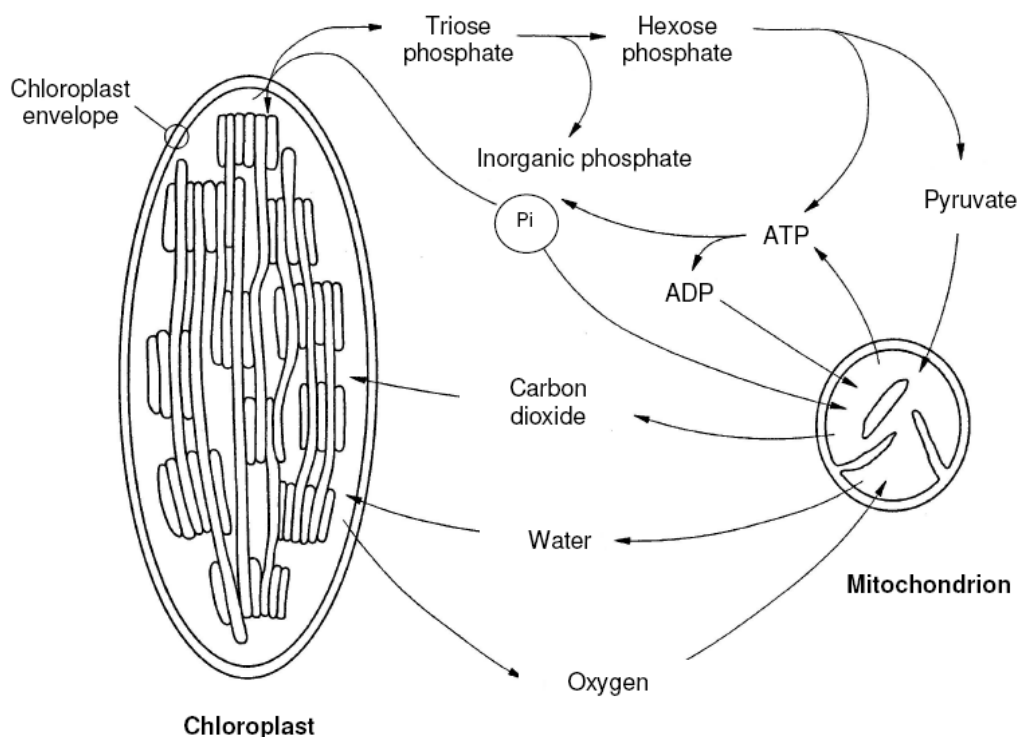


Fig. 6.1

(a) State two structural features of chloroplasts and mitochondria that are also present in prokaryotic cells.

.....
[1]

(b) A leafy shoot is sealed inside a transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

With reference to **Fig. 6.1**,

(i) Explain how it is possible for the oxygen concentration to remain constant.

.....

[3]

(ii) Explain why there is no build up in the concentration of inorganic phosphate ions (P_i) inside the mitochondria as a result of the inward passage of P_i .

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[1½]

(c) An investigation was carried out to find out the effect of three inhibitors, **A**, **B** and **C**, on the electron transport chain in the mitochondria. In each of three experiments, a different inhibitor was added. The table below shows the state of the electron carriers, **W–Z**, after the addition of inhibitor.

Inhibitor added	Electron carrier			
	W	X	Y	Z
A	oxidised	reduced	reduced	oxidised
B	oxidised	oxidised	reduced	oxidised
C	reduced	reduced	reduced	oxidised

(i) Give the order of the electron carriers in this electron transport chain.

Order [1]

(ii) Explain your answer.

.....
[1]

(d) A student investigated the effects of temperature and light intensity on the rate of photosynthesis of an aquatic plant.

Fig. 6.2 shows the results of the investigation.

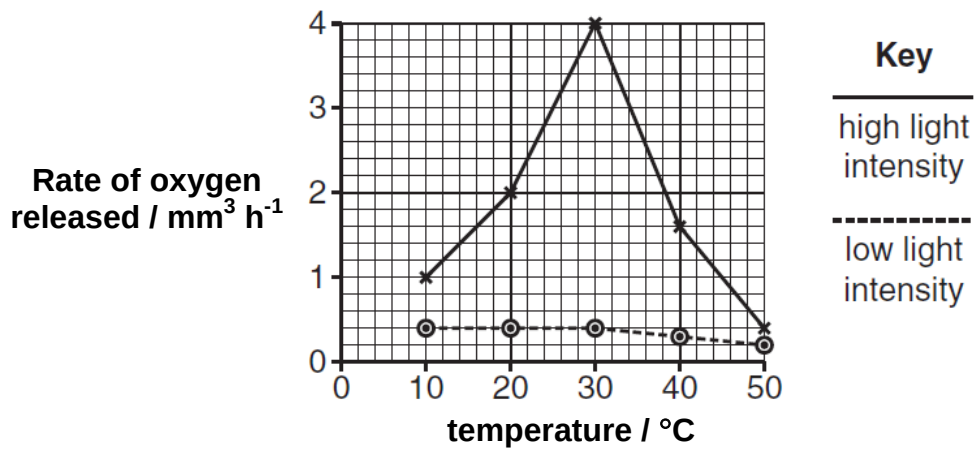


Fig 6.2

With reference to Fig. 6.2,

(i) describe the results of the investigation.

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[2]

(ii) suggest explanations for the results for high light intensity above 30°C.

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[1½]

[Q6 Total: 11]

7. A mammoth can be any species belonging to the extinct genus *Mammuthus*, proboscideans commonly equipped with long, curved tusks. They were members of the family Elephantidae which contains, along with mammoths, the two genera of modern elephants and their ancestors.

About three million years ago, the ancestors of mammoths migrated from Africa into Europe and Asia. There, about 1.7 million years ago, the steppe mammoth (*Mammuthus armeniacus*) evolved and became adapted to the cooler conditions. Then, about 700 000 years ago, as the climate changed and the Arctic became much colder, the woolly mammoth (*Mammuthus primigenius*) evolved. Woolly mammoths showed a number of characteristics that include thick fur, small ears and small tails.

(a) The term *species* is often used in the context of evolution of new species. Explain the meaning of the term *species*.

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.....[3]

(b) Explain how variation and natural selection may have brought about the evolution of the woolly mammoth from the steppe mammoth.

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.....[4]

(c) A frozen, 43 000 year old woolly mammoth was found in Western Siberia. Its DNA was extracted and sequenced. The phylogenetic relationship of the woolly mammoth, Asian elephant (*Elephas maximus*) and African elephant (*Loxodonta africana*) was then investigated.

(i) Explain how molecular methods can be used to elucidate the evolutionary relationships of the three members of the family Elephantidae.

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.....[2]

(ii) Explain why molecular evidence is better than morphological evidence in determining phylogeny.

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.....[3]

[Q7 Total: 12]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8** **(a)** Explain the importance of mitosis and the need for fine control of replication. [6]
- (b)** Outline how the structure of cellulose is related to its specific function. [5]
- (c)** Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [9]

[Total: 20]

OR

- 9** **(a)** Describe and explain the transmission of an action potential along a myelinated neurone. [8]
- (b)** Explain how a cholinergic synapse functions. [8]
- (c)** Contrast nervous and hormonal control. [4]

[Total: 20]